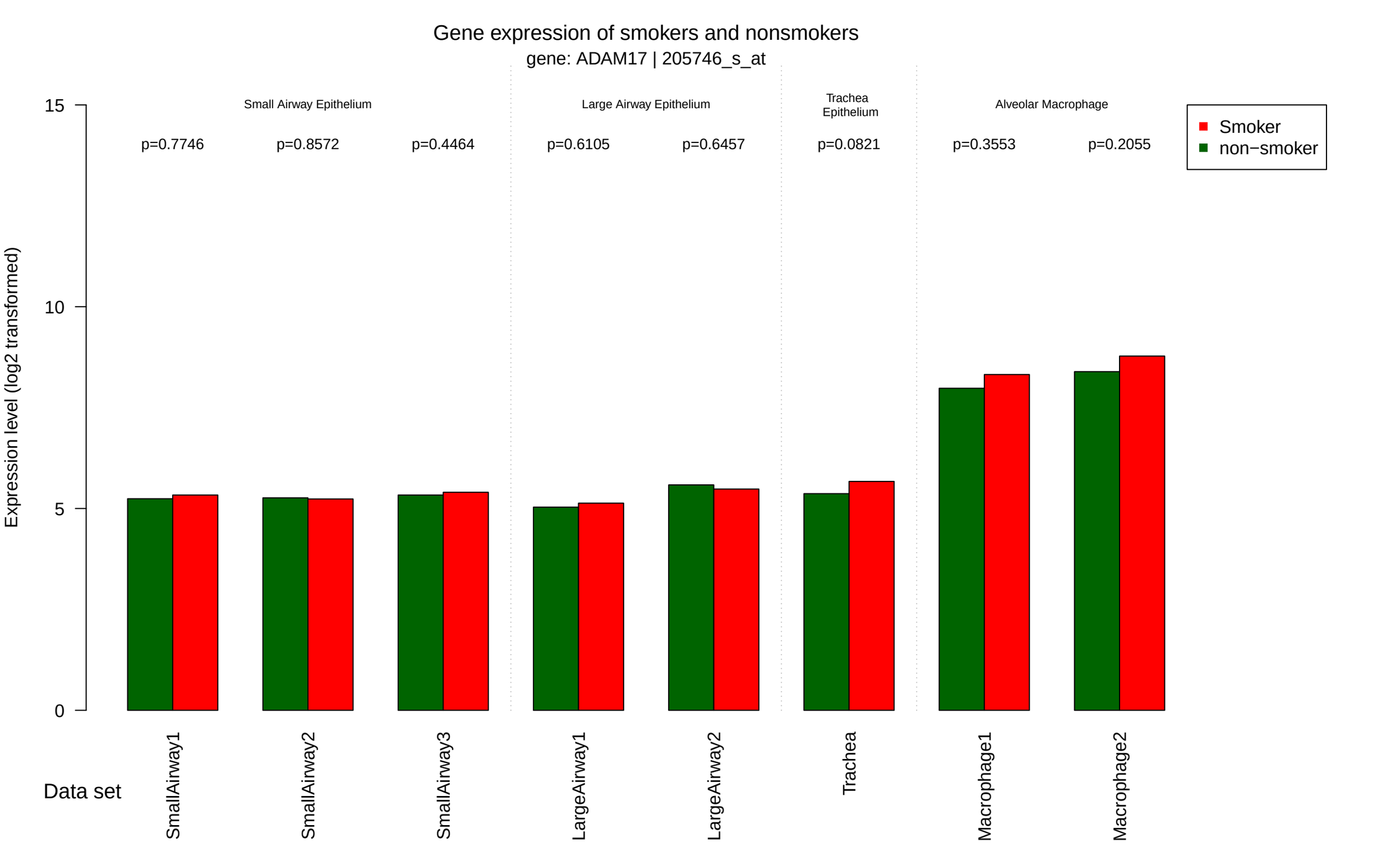
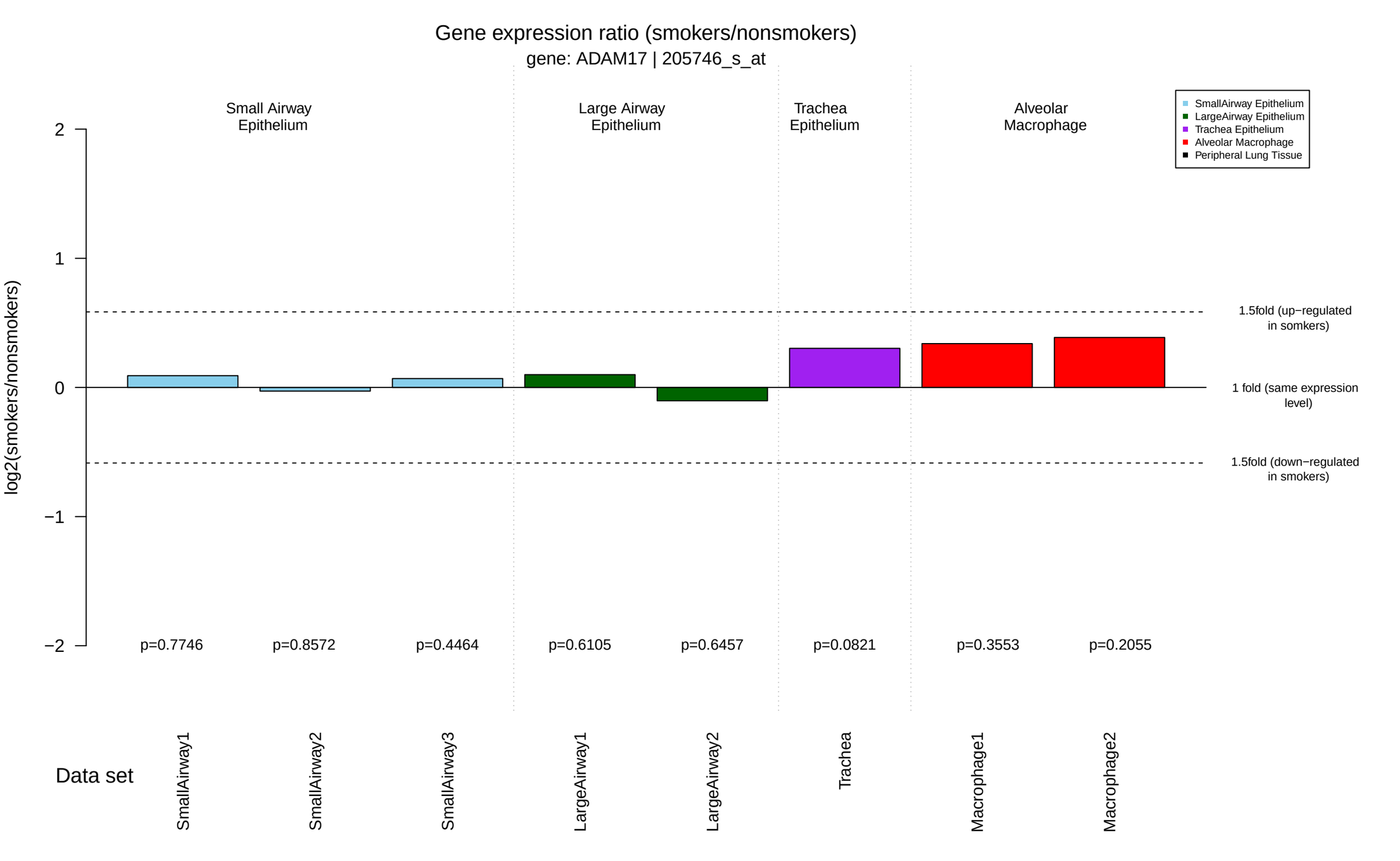
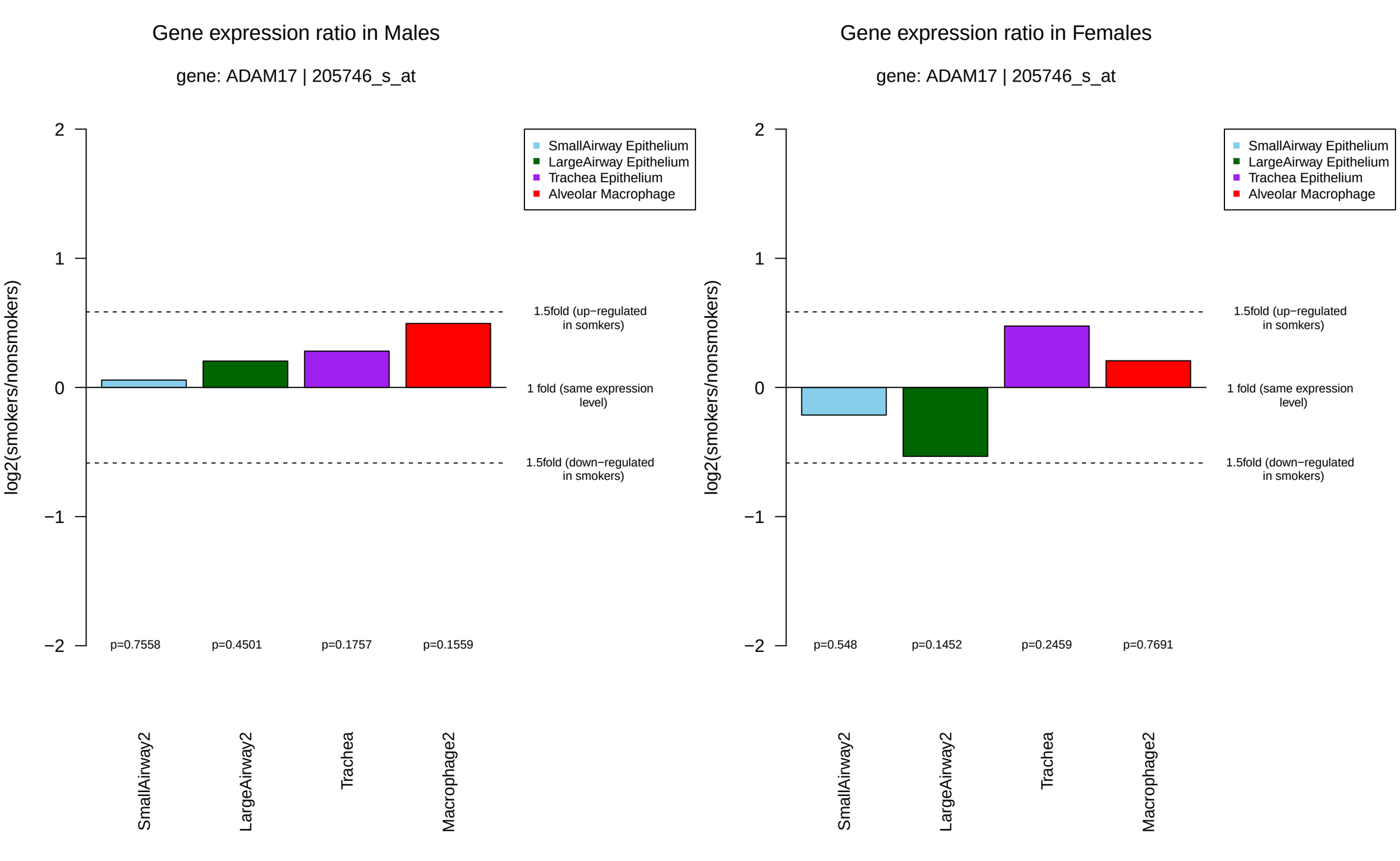
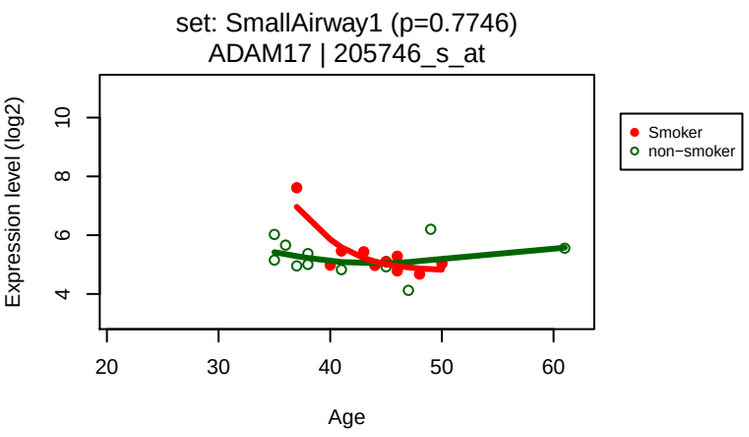
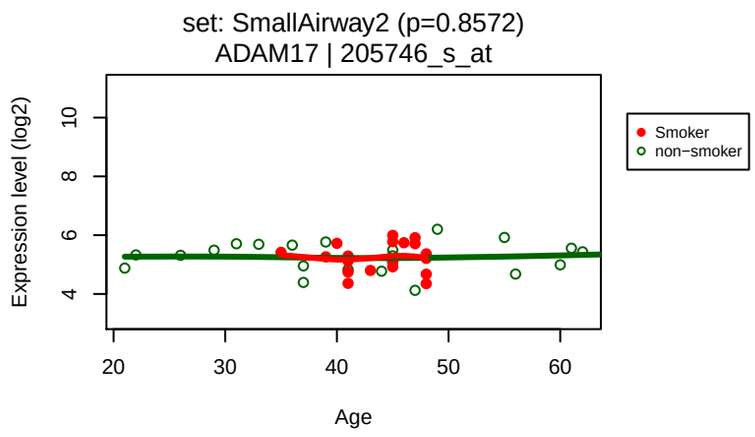
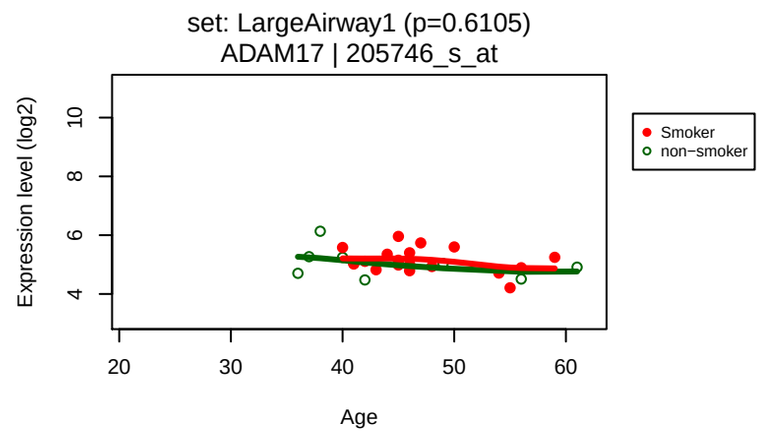

Gene expression of smokers and nonsmokers
gene: ADAM17 | 205746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7746
p=0.8572
p=0.4464
p=0.6105
p=0.6457
p=0.0821
p=0.3553
p=0.2055
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

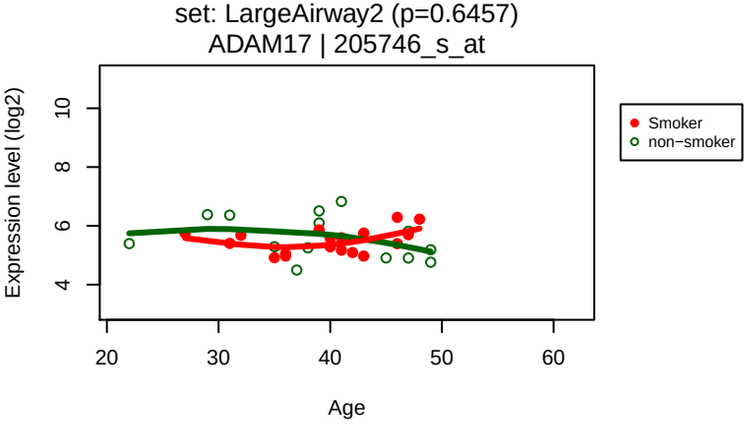

Gene expression ratio (smokers/nonsmokers)
gene: ADAM17 | 205746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7746
p=0.8572
p=0.4464
p=0.6105
p=0.6457
p=0.0821
p=0.3553
p=0.2055
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

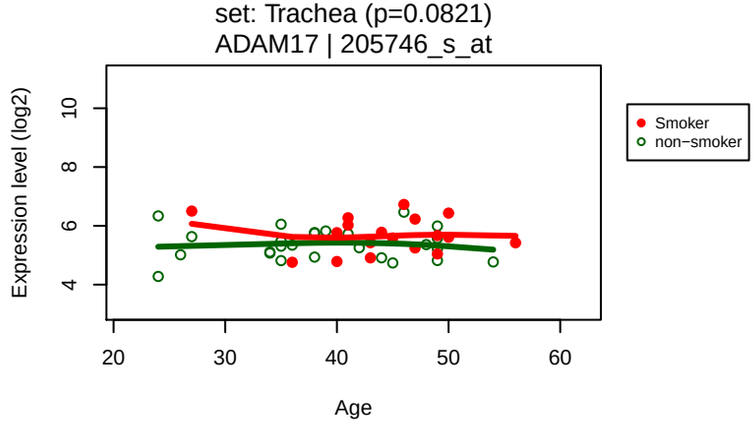

Gene expression ratio in Males
gene: ADAM17 | 205746_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7558
p=0.4501
p=0.1757
p=0.1559
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADAM17 | 205746_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.548
p=0.1452
p=0.2459
p=0.7691
SmallAirway2
LargeAirway2
Trachea
Macrophage2

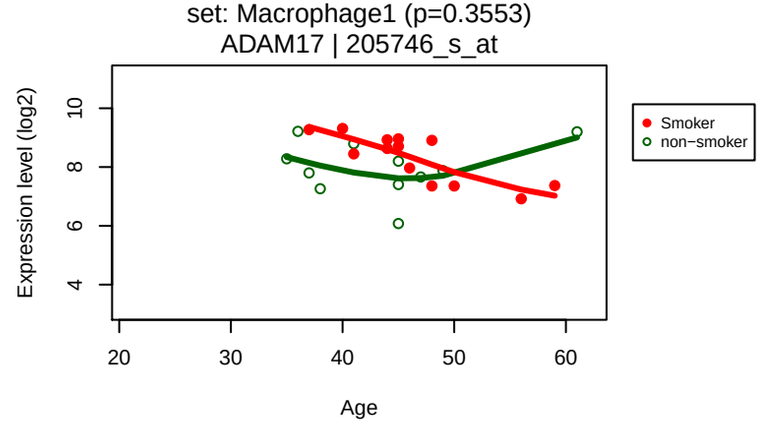

set: SmallAirway1 (p=0.7746)
ADAM17 | 205746_s_at
Expression level (log2)
Age
Smoker
non−smoker

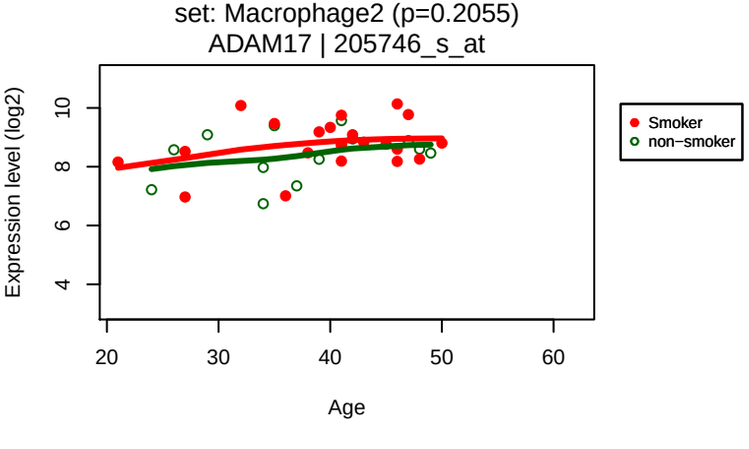

set: SmallAirway2 (p=0.8572)
ADAM17 | 205746_s_at
Expression level (log2)
Age
Smoker
non−smoker

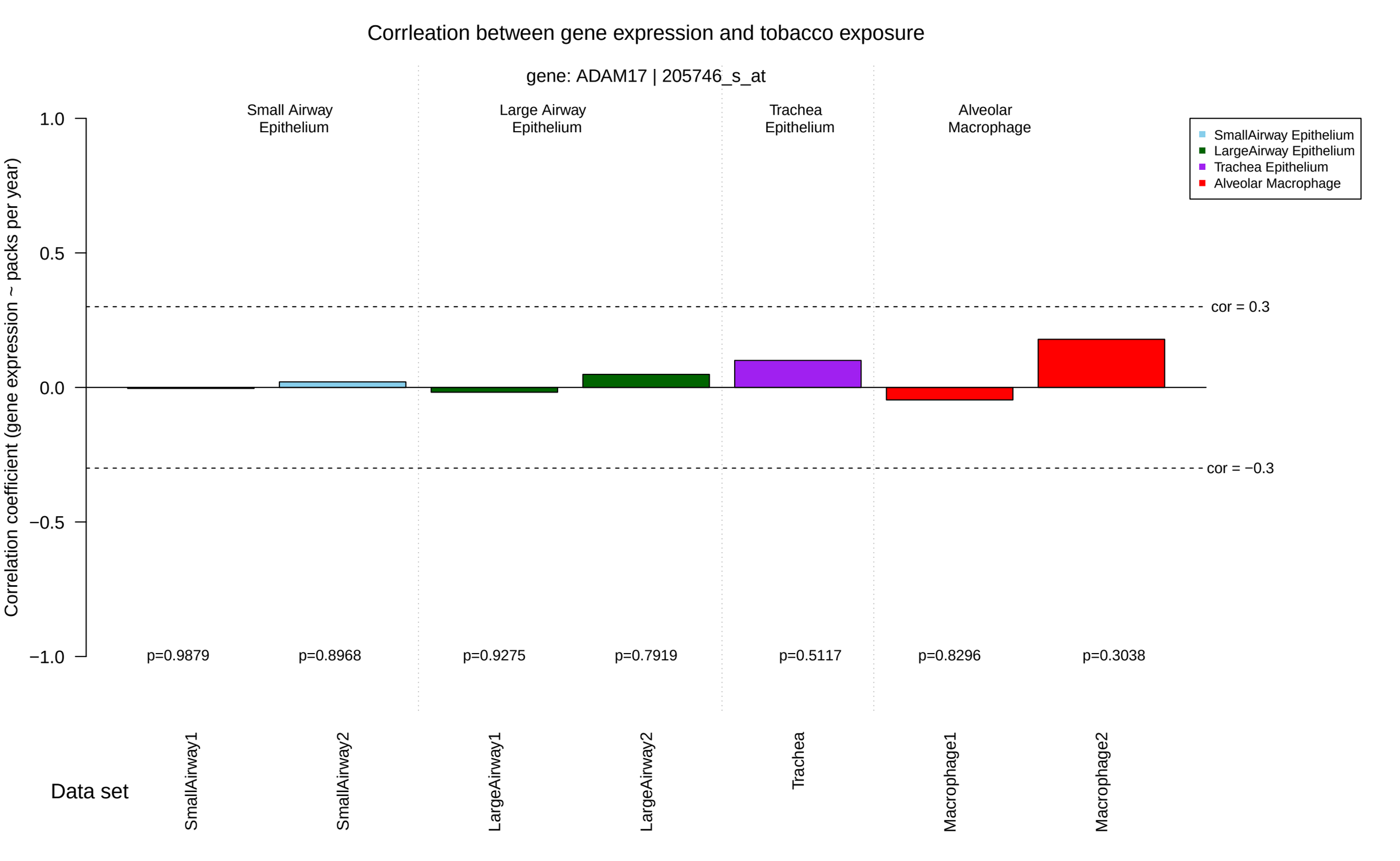

set: LargeAirway1 (p=0.6105)
ADAM17 | 205746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6457)
ADAM17 | 205746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0821)
ADAM17 | 205746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3553)
ADAM17 | 205746_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2055)
ADAM17 | 205746_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADAM17 | 205746_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9879
p=0.8968
p=0.9275
p=0.7919
p=0.5117
p=0.8296
p=0.3038
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2